Stability Protocol STP-204-003

Product: BRX-204 (orvatimab) 150 mg/mL solution for injection in 1 mL prefilled syringe

Protocol date: 20 June 2022

Prepared by: Biologics Stability Group

1. Purpose

This protocol defines the stability program supporting the registration and commercial shelf life of BRX-204 drug product in the prefilled syringe presentation, in accordance with ICH Q5C and ICH Q1A(R2).

2. Batches

Three primary batches manufactured at commercial scale (Lots OV-C21, OV-C22, OV-C23) using drug substance from three independent drug substance lots.

3. Storage conditions and timepoints

Condition	Orientation	Timepoints (months)
5°C ± 3°C (long-term)	Horizontal	0, 1, 3, 6, 9, 12, 18, 24, 30, 36
25°C ± 2°C / 60% RH (accelerated)	Horizontal	0, 1, 2, 3, 6
40°C ± 2°C / 75% RH (stress)	Horizontal	0, 0.5, 1

Syringes are stored horizontally so that the solution is in contact with the plunger stopper, representing the worst case for leachables.

4. Tests and acceptance criteria

Test	Acceptance criterion
Potency (cell-based assay)	70–130% of reference standard
Purity, SEC-HPLC	Main peak NLT 95.0%; HMW species NMT 3.0%
Charge variants, CEX	Acidic variants NMT 35%
Subvisible particles	≥10 µm: NMT 6000 per container; ≥25 µm: NMT 600 per container
Polysorbate 80 content	0.10–0.30 mg/mL
Container closure integrity	No leak detected (HVLD)

5. Bracketing and matrixing

A matrixing design is applied to the accelerated condition only: at the 2-month timepoint, two of the three batches are tested. Matrixing is justified by the low variability observed in development stability data (relative standard deviation of HMW species below 5%) and is consistent with ICH Q1D. No bracketing is applied, as only one fill volume and one container size are registered.

6. Deviations

Storage excursions are managed per SOP-QA-021. Any excursion affecting these samples shall be cross-referenced in the stability report.

7. Post-approval commitment

Borealis Therapeutics commits to place the first three commercial production batches on long-term stability at 5°C through 36 months, and to add at least one batch per year to the ongoing stability program.